

Baggage Alarms Dashboard

Overview

An equipment fault monitoring and alert management dashboard. Tracks system alarms generated across baggage handling equipment (conveyors, scanners, sorters, etc.) to support maintenance prioritisation and operational efficiency.

Key Metrics

Metric	Value	Definition
Total Alarms	412,055	System alerts over measurement period
Avg Daily Alarms	4,479	Daily average; volatility shown with moving average
Avg Hourly Alarms	17,169	Hourly average across all times
Avg Alarms (15 min)	4,292	Granular time-bucket average
Avg Alarms (5 min)	1,431	Sub-hour granularity for peak detection
Peak Time Bucket	07:30–07:45	15-minute period with highest alarms

Dashboard Pages

Daily Alarms (Main Page): Total alarm count, daily/hourly/15-min/5-min averages. Trend chart with moving average overlay to identify seasonal patterns and spikes.

Alarms Over Time (Performance Analysis): Multi-resolution view (60-minute, 15-minute, 5-minute). Shows alarm count and fault resolution time (MTTR) by time bucket. Peak hour selector to zoom into critical periods.

Alarms By Equipment (Equipment Analysis): Pareto chart of alarms by equipment area (Transport, Unknown, Transfer Bag Acquisition, Screening, etc.). Shows which 20% of equipment generates 80% of alarms. Adjustable Pareto threshold.

Equipment Fault Resolution & Incidents (Fault Resolution Time): Equipment type and area breakdown. Tracks incident counts and average MTTR. Identifies problem areas requiring maintenance focus.

Use Cases

- **Maintenance Prioritisation:** Pareto analysis identifies which equipment is causing most disruption; focus maintenance resources on the vital few.
- **Fault Trend Analysis:** Detect seasonal patterns, equipment degradation trends, or post-maintenance improvement.
- **SLA Tracking:** Monitor MTTR by equipment type to ensure maintenance SLAs are met.
- **Capacity Planning:** Peak hour analysis informs staffing and maintenance scheduling.

Technical Stack

Component	Technology
Visualisation	Power BI Desktop / Service
Data Source	Baggage equipment fault log system
Refresh Frequency	Daily (23/10/25 last update)
Key Filters	Date, Hour/15 Min/5 Min bucket, Equipment Type/Zone/Area